

UNDERKHAAN, Mongolia

WMO: 443040

Lat: 47.304N

Lon: 110.608E

Elev: 1033

StdP: 89.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.7	(c) -33.4	(d) -38.5	(e) 0.1	(f) -35.6	(g) -36.2	(h) 0.1	(i) -33.3	(j) 11.9	(k) -16.2	(l) 10.2	(m) -15.7	(n) 3.0	(o) 230	(p) 0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 32.1	(d) 16.7	(e) 29.8	(f) 15.8	(g) 27.8	(h) 15.3	(i) 18.7	(j) 26.7	(k) 17.7	(l) 25.4	(m) 16.9	(n) 24.4	(o) 5.5	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.2	20.6	15.3	12.4	20.1	14.4	11.6	19.8	57.6	26.8	54.5	25.4	51.7	24.4	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.6	9.7		-38.1	35.3	2.5	2.3	-39.9	37.0	-41.4	38.3	-42.8	39.6	-44.6	41.2
				-38.0	20.7	2.5	1.1	-39.8	21.5	-41.2	22.1	-42.6	22.7	-44.4	23.5
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-23.0	-18.0	-7.3	3.8	11.1	17.6	20.5	18.1	11.4	1.7	-11.4	-20.4		
	DBStd	16.09	5.22	6.13	6.94	4.87	4.74	3.78	3.22	3.46	4.62	5.15	7.31	5.17		
	HDD10.0	4466	1024	786	537	192	43	1	0	0	37	260	643	943		
	HDD18.3	6698	1283	1019	795	434	227	58	13	47	212	517	893	1201		
	CDD10.0	971	0	0	0	8	78	229	325	250	79	2	0	0		
	CDD18.3	161	0	0	0	0	4	36	79	38	4	0	0	0		
	CDH23.3	2058	0	0	0	5	142	526	855	453	78	0	0	0		
	CDH26.7	708	0	0	0	1	38	184	323	149	13	0	0	0		
(14) Wind	WSAvg	3.9	3.6	3.7	4.1	5.1	5.3	4.1	3.4	3.2	3.8	3.8	3.6	3.6		
(15) Precipitation	PrecAvg	243	2	3	3	7	17	49	66	57	25	9	3	3		
	PrecMax	411	9	12	14	28	41	164	287	194	86	38	13	34		
	PrecMin	123	0	0	0	0	3	7	2	7	0	0	0	0		
	PrecStd	65	2	3	3	7	10	31	47	40	19	9	4	5		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.0	2.2	14.2	23.1	30.8	34.0	35.8	33.7	28.4	20.1	8.1	-4.2		
		MCWB	-7.9	-2.1	4.1	8.9	13.2	15.6	17.8	16.9	13.6	8.7	1.9	-6.3		
	2%	DB	-9.9	-1.8	9.7	19.1	27.1	31.1	32.6	30.4	25.5	16.3	4.7	-7.9		
		MCWB	-11.2	-4.9	2.1	7.3	12.0	14.9	17.7	16.4	13.0	6.8	-0.1	-9.3		
	5%	DB	-12.5	-5.1	6.3	16.4	24.3	28.7	30.0	27.9	23.1	13.2	1.7	-10.1		
		MCWB	-13.6	-7.3	0.2	6.0	10.8	14.6	16.9	15.8	12.0	5.3	-2.1	-11.3		
	10%	DB	-14.8	-8.0	3.3	13.7	21.7	26.2	27.9	25.7	20.4	10.6	-0.9	-12.3		
		MCWB	-15.6	-9.5	-1.4	4.7	9.7	13.9	16.6	15.2	11.0	3.9	-3.7	-13.2		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.8	-1.9	4.5	9.5	14.1	18.1	20.6	19.5	16.1	9.1	2.1	-6.1		
		MCDB	-6.2	2.1	13.7	20.3	26.7	27.5	29.7	27.2	23.2	18.3	7.7	-4.1		
	2%	WB	-11.2	-4.8	2.2	7.8	12.6	16.8	19.2	18.2	14.6	7.3	0.0	-9.2		
		MCDB	-9.9	-1.9	9.0	17.9	25.1	26.3	27.9	25.1	21.2	15.2	4.5	-8.0		
	5%	WB	-13.5	-7.2	0.3	6.2	11.3	15.8	18.3	17.3	13.3	5.6	-1.9	-11.2		
		MCDB	-12.6	-5.1	6.0	15.7	22.7	24.7	26.4	24.3	20.2	12.6	1.4	-10.2		
	10%	WB	-15.6	-9.5	-1.4	4.7	10.0	15.0	17.5	16.4	12.0	4.1	-3.7	-13.2		
		MCDB	-14.9	-8.0	3.0	13.1	20.2	23.8	25.1	23.2	19.3	10.2	-1.0	-12.3		
(36) Mean Daily Temperature Range	5% DB	MDBR	12.4	15.2	15.0	14.9	15.5	13.5	12.3	12.9	14.3	13.8	12.8	11.7		
		MCDBR	13.9	17.1	18.1	19.4	20.5	17.7	16.1	16.8	18.0	17.9	14.7	13.0		
	5% WB	MCWBR	12.8	14.4	11.7	10.7	9.3	6.9	5.3	6.6	8.3	10.5	10.7	11.7		
		MCWBR	13.9	17.0	17.3	18.1	18.3	14.2	13.0	13.1	14.6	16.8	14.6	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.271	0.309	0.364	0.416	0.431	0.396	0.371	0.331	0.293	0.262	0.247		
	taud		2.338	2.299	2.270	2.173	2.060	2.130	2.302	2.342	2.389	2.427	2.390	2.326		
	Ebn at Noon		856	902	912	886	848	835	862	871	879	866	828	806		
	Edh at Noon		70	92	112	137	159	150	125	115	99	81	67	63		
(44) All-Sky Solar Radiation	RadAvg		1.96	3.10	4.63	5.83	6.63	6.53	6.26	5.70	4.66	3.29	2.11	1.57		
	RadStd		0.13	0.20	0.18	0.21	0.30	0.40	0.28	0.29	0.27	0.19	0.12	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page